

Near-Dirac-Delta Probability Density Functions

A Multidisciplinary Perspective from Mathematics, Statistics, Economics, Finance, Political Science, Warfare, and Artificial Intelligence

Soumadeep Ghosh

Kolkata, India

Abstract

The Dirac delta distribution occupies a central role in mathematics, physics, statistics, economics, finance, engineering, military operations research, artificial intelligence, and data science. Since the Dirac delta is not an ordinary probability density function, practical applications employ sequences of ordinary densities that converge to it. Such approximations, often called near-Dirac-delta densities or delta sequences, model increasingly concentrated uncertainty around a target point. This paper surveys major examples, proves their convergence, and discusses applications across multiple disciplines.

The paper ends with “The End”

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1 Introduction

The Dirac delta centered at a is denoted by $\delta(x - a)$ and satisfies

$$\int_{-\infty}^{\infty} f(x)\delta(x - a) dx = f(a).$$

Although δ is not a conventional function, many ordinary probability densities approach it in a suitable limiting sense.

Such approximations arise in:

- Bayesian statistics.
- Signal processing.
- Economic equilibrium models.
- Financial limit-order execution.
- Political forecasting.
- Military targeting and weapons systems.
- Machine learning and kernel methods.
- Reinforcement learning.

2 Mathematical Foundations

Definition 1 (Delta Sequence). *A family of densities $\{f_\varepsilon\}$ is a delta sequence at a if*

$$f_\varepsilon(x) \geq 0,$$

$$\int_{-\infty}^{\infty} f_\varepsilon(x) dx = 1,$$

and

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} \phi(x) f_\varepsilon(x) dx = \phi(a)$$

for every continuous test function ϕ .

3 Examples of Near-Dirac-Delta Densities

3.1 Gaussian

$$f_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-a)^2}{2\sigma^2}\right).$$

3.2 Uniform

$$f_\varepsilon(x) = \begin{cases} \frac{1}{2\varepsilon}, & |x-a| < \varepsilon, \\ 0, & \text{otherwise.} \end{cases}$$

3.3 Laplace

$$f_b(x) = \frac{1}{2b} e^{-|x-a|/b}.$$

3.4 Cauchy

$$f_\varepsilon(x) = \frac{1}{\pi} \frac{\varepsilon}{(x-a)^2 + \varepsilon^2}.$$

3.5 Triangular

$$f_\varepsilon(x) = \begin{cases} \frac{1}{\varepsilon} \left(1 - \frac{|x-a|}{\varepsilon}\right), & |x-a| < \varepsilon, \\ 0, & \text{otherwise.} \end{cases}$$

4 Comparison Table

Family	Support	Integral	Limit Parameter
Gaussian	$(-\infty, \infty)$	1	$\sigma \rightarrow 0$
Uniform	$[a - \varepsilon, a + \varepsilon]$	1	$\varepsilon \rightarrow 0$
Laplace	$(-\infty, \infty)$	1	$b \rightarrow 0$
Cauchy	$(-\infty, \infty)$	1	$\varepsilon \rightarrow 0$
Triangular	$[a - \varepsilon, a + \varepsilon]$	1	$\varepsilon \rightarrow 0$

Table 1: Representative delta-sequence families

5 Convergence Proof

Theorem 1. *The Gaussian family converges to $\delta(x-a)$ as $\sigma \rightarrow 0$.*

Proof. Consider

$$I_\sigma = \int_{-\infty}^{\infty} \phi(x) \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-a)^2/(2\sigma^2)} dx.$$

Substitute

$$u = \frac{x-a}{\sigma}.$$

Then

$$I_\sigma = \int_{-\infty}^{\infty} \phi(a + \sigma u) \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du.$$

As $\sigma \rightarrow 0$,

$$\phi(a + \sigma u) \rightarrow \phi(a).$$

Dominated convergence yields

$$I_\sigma \rightarrow \phi(a) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = \phi(a).$$

Hence

$$f_\sigma \Rightarrow \delta(x-a).$$

□

6 Statistical Interpretation

Suppose

$$X_n \sim N\left(a, \frac{1}{n}\right).$$

Then

$$\text{Var}(X_n) = \frac{1}{n} \rightarrow 0.$$

By Chebyshev's inequality,

$$P(|X_n - a| > \eta) \leq \frac{1}{n\eta^2}.$$

Therefore

$$X_n \xrightarrow{P} a.$$

7 Economic Interpretation

Consider a market equilibrium price P^* .

Let

$$P_n \sim N(P^*, \sigma_n^2)$$

with

$$\sigma_n^2 \rightarrow 0.$$

Then the distribution of observed prices approaches

$$\delta(P - P^*).$$

This corresponds to a frictionless perfectly informed market.

7.1 Welfare Interpretation

Consumer surplus:

$$CS = \int_0^{Q^*} D(q) dq - P^* Q^*.$$

Producer surplus:

$$PS = P^* Q^* - \int_0^{Q^*} S(q) dq.$$

As uncertainty vanishes, equilibrium outcomes become deterministic.

8 Finance

Suppose a risky asset price satisfies

$$S_T \sim N(\mu, \sigma^2).$$

As

$$\sigma \rightarrow 0,$$

the terminal value approaches

$$\delta(S_T - \mu).$$

Option valuation becomes

$$C = e^{-rT} \max(\mu - K, 0).$$

Randomness disappears and valuation becomes deterministic.

9 Political Science and Geopolitics

Election forecasting often produces a probability density for vote share V .

As polling uncertainty decreases,

$$f(V) \rightarrow \delta(V - V^*).$$

Similarly, in geopolitical forecasting, beliefs about conflict outcomes may evolve from broad uncertainty toward concentrated expectations.

10 Warfare Economics

Let targeting error be modeled by

$$E \sim N(0, \sigma^2).$$

As weapons guidance improves,

$$\sigma \rightarrow 0.$$

Then

$$f(E) \rightarrow \delta(E).$$

Expected collateral damage decreases while precision increases.

11 Artificial Intelligence and Machine Learning

11.1 Kernel Density Estimation

Kernel density estimation uses

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right).$$

As

$$h \rightarrow 0,$$

each kernel behaves like a near-Dirac-delta density.

11.2 Attention Mechanisms

Transformer attention weights satisfy

$$\alpha_i = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

As temperature tends to zero,

$$\alpha_i \rightarrow \begin{cases} 1, & i = i^*, \\ 0, & i \neq i^*. \end{cases}$$

The distribution approaches a discrete analogue of a Dirac delta.

12 Algorithm

Delta-Sequence Simulation

1. Choose center a .
2. Choose family type.
3. Select concentration parameter.
4. Generate random samples.
5. Reduce parameter toward zero.
6. Observe convergence.

Computational complexity:

$$O(n).$$

13 Logic and Philosophy

A near-Dirac-delta density formalizes the distinction between

- certainty,
- near certainty,
- probabilistic belief.

Epistemically,

$$P(X = a) = 1$$

corresponds to certainty.

Near-delta distributions model beliefs that are highly concentrated without being logically certain.

14 Illustration Using TikZ

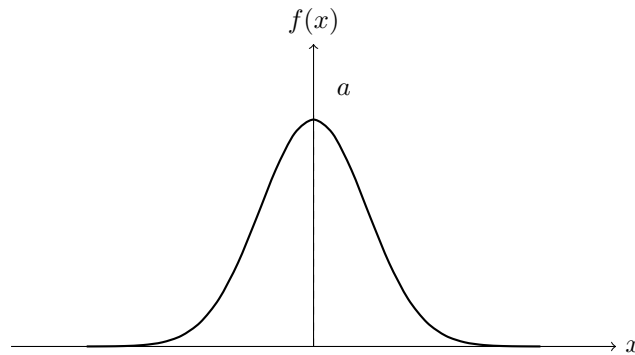


Figure 1: A highly concentrated Gaussian approaching a Dirac delta

15 Conclusion

Near-Dirac-delta densities provide a unifying mathematical language for concentrated uncertainty. They appear naturally across statistics, economics, finance, political science, geopolitics, military operations research, machine learning, and philosophy. The convergence of ordinary probability densities toward the Dirac delta allows deterministic models to emerge as limiting cases of stochastic systems.

References

- [1] W. Rudin, *Real and Complex Analysis*, McGraw–Hill.
- [2] W. Feller, *An Introduction to Probability Theory and Its Applications*, Wiley.
- [3] G. Casella and R. Berger, *Statistical Inference*, Duxbury.
- [4] S. Haykin, *Neural Networks and Learning Machines*, Pearson.
- [5] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
- [6] H. Varian, *Microeconomic Analysis*, W. W. Norton.
- [7] M. Osborne and A. Rubinstein, *A Course in Game Theory*, MIT Press.
- [8] C. von Clausewitz, *On War*.
- [9] K. Waltz, *Theory of International Politics*.

Glossary

Dirac Delta

Generalized function satisfying $\int f(x)\delta(x-a)dx = f(a)$.

Delta Sequence

Family of ordinary densities converging to a Dirac delta.

Weak Convergence

Convergence against all continuous test functions.

Kernel

Localized weighting function used in density estimation.

Equilibrium

State in which economic forces balance.

Attention

Probability distribution over tokens in a transformer.

Targeting Error

Deviation between intended and realized strike location.

The End